

Chapter 2: Physical layer

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Overview

- Physical layer is responsible for transmission of a stream of bits
 - Put bits from a machine to a medium
 - Pick bits from the medium give to receiver
- Some issues
 - Medium
 - Line Encoding: representing the digital logic levels using the physical attributes associated with the media.
 - Multiplexing

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From signal to packet

Analog Signal



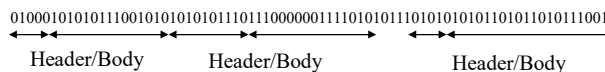
“Digital” Signal



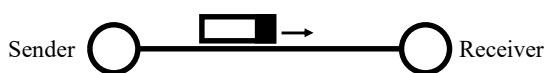
Bit Stream

0 0 1 0 1 1 1 0 0 0 1

Packets

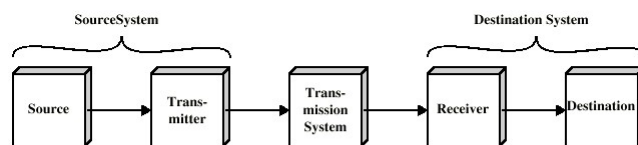


Packet
Transmission

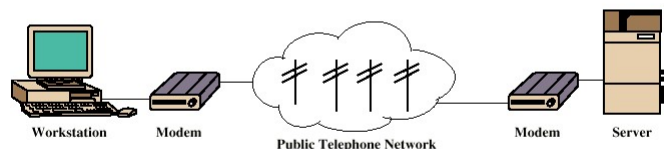


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Model of data transmission system



(a) General block diagram



(b) Example

Figure 1.1 Simplified Communications Model

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Data Communication networks

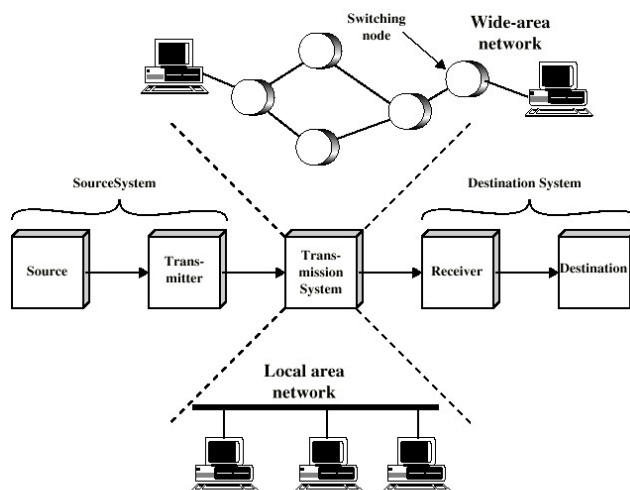


Figure 1.3 Simplified Network Models



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Communication Media

- Wired media: cable guides signals
 - Twisted Pair
 - Coaxial Cable
 - Fiber Optics
- Wireless media: no cable guides signal
 - Radio
 - Infra red
 - Light
 - ...

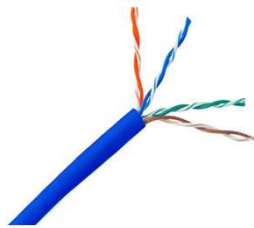


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Twisted pair



(a)



(b)

(a) Category 3 UTP.

(b) Category 5 UTP.

Twisted pair

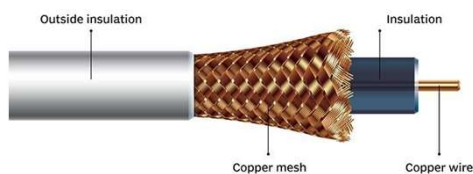
- Contains several pairs of copper, cable in the one pair is twisted together.
- Two kinds of twisted pair:
 - STP-Shielded Twisted Pair:
 - There is a metal coat, not popular
 - UTP-Unshielded Twisted Pair:
 - No metal coat, popular

Evaluation

- Cheap, simple
- Widely used
- Weak resistance to noise
- Short Transmission distance
- Need amplification after each 5km in analog transmission
- In digital transmission
 - Need repeater after each 2 km
- In Ethernet LAN deployment < 100m
- Limited speed (100Mbps)

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Coaxial cable



Category	Impedance	Use
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

Structure:

- Inner conduct is coated by an insulator environment
- Shielded by a metal grill
- A plastic cover for protection.

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Application

- Using in TV transmission
 - For transmission of telephone signal
 - 10,000 calls in the same time
 - Is being replaced by fiber optics
 - Linking the computers of the short distance
 - LAN 10BaseT, 100BaseT, ...
- For digital transmission
 - Repeater should be used after each 1km
 - More repeater is needed for high speed transmission

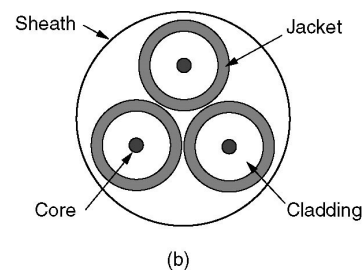
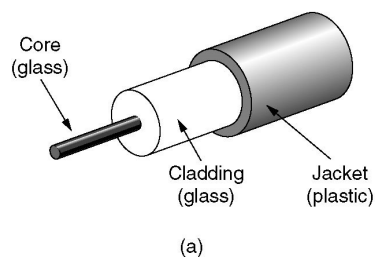


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Optical fiber



- (a) Single core
(b) Cable with 3 cores

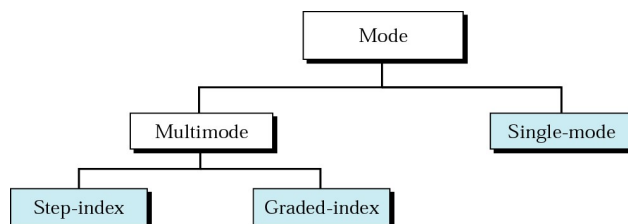
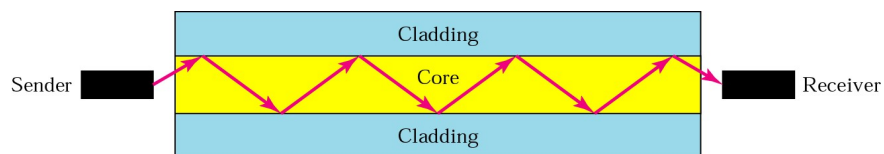


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Optical fiber transmission mode

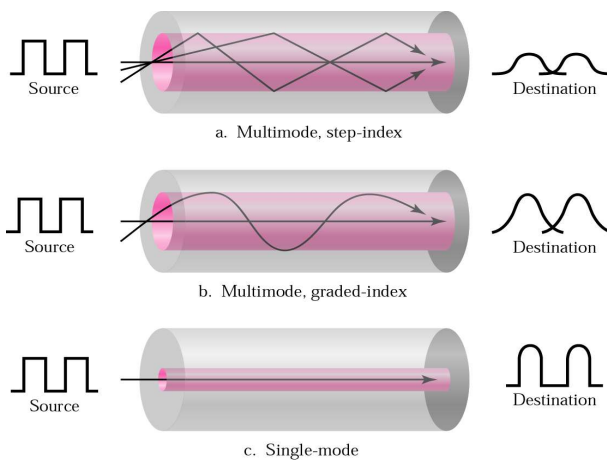


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Optical fiber



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Optical fiber

- Multimode stepped index :
 - Several beam travel in slightly different direction
 - Beams arrive in different delay
 - Pulse can easily distort
- Multimode graded index:
 - Index reduce gradually from the center to cladding.
 - Beams closed to center travel slower than cladding. Beams travel in curve form.
 - Reduce pulse distortion.

Optical fiber

- Single mode:
 - Index change less from center to cladding in comparison with multimode.
 - Beams travel along the center axe.
 - Pulses experience less distortion.

Application of optical fiber

- Used for long distance transmission
- Used for communication in metropolitan networks
- Used for connecting routers of ISP
- Used in backbone part of a LAN
- Advantage in comparison with other cables
 - Large data rate
 - Small and light cable
 - Low attenuation
 - Better isolation from electromagnetic environment
 - Large distance between repeaters
 - Multimode → 10km
 - Singlemode → 40 km

Wireless media

- Terrestrial microwave
 - Used for metropolitan connection, for cellular network
- Microwave satellite
 - Used in TV, Long distance telephone communication
- Radio broadcast
- Infrared
 - Small scope, low data rate, unable to travel through the wall

Wireless media

- Radio wave:
 - Wavelengths: 1mm – 100.000km
 - Frequencies: 3 Hz – 300 GHz
 - Ex: Bluetooth, WIFI
- Microwave:
 - Wavelengths: 1mm-1 m
 - Frequencies: 300 MHz-300 GHz
 - Terrestrial microwave : metro connection, cellular communication
 - Satellite microwave: TV, long distance telephone



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Wireless media

- Infra red:
 - Wavelengths: 700 nm- 1 mm
 - frequency: 300 GHz-430 THz
 - Small scope, no wall penetration
 - Ex: use in remote controls
- Free Space Optics
 - Wavelengths: 850nm, 1300nm, 1550 nm

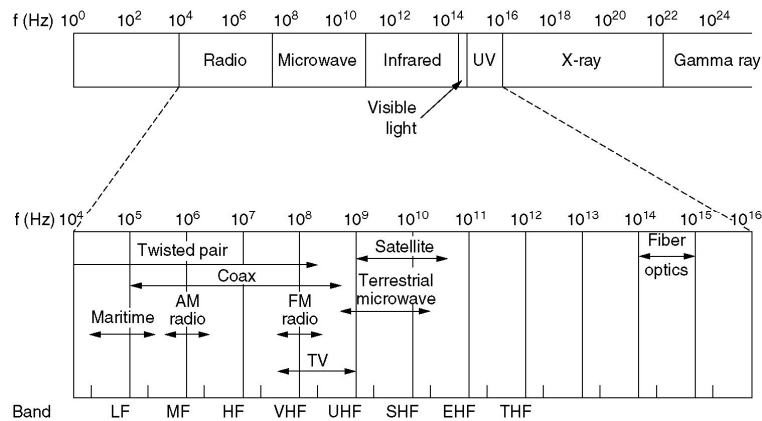


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Frequency range of transmission channels



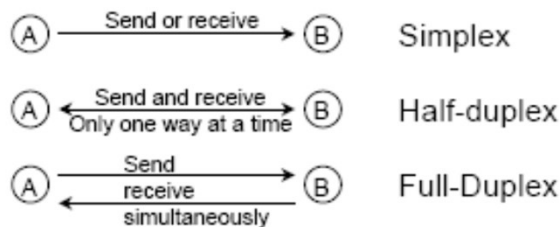
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Transmission methods

- Simplex: Data is transmitted in one direction
- Full Duplex: Data can be transmitted in both directions at the same time
- Half duplex: Data can be transmitted in both directions but one direction at a time.



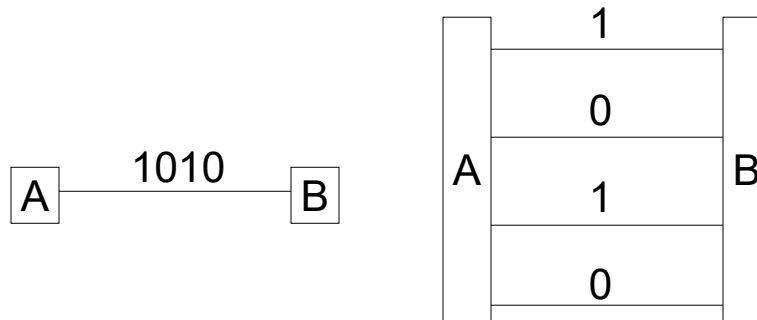
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Transmission format

- Sequent transmission: Transmit 1 bit at a moment (over a signal line)
- Parallel transmission: Transmit multiple bits at the same time (over multiple signal lines)



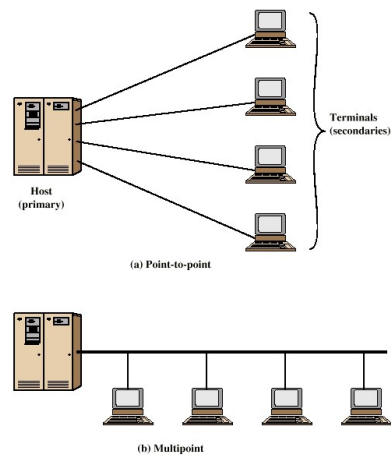
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Topology

- Point-to-point
 - Star
 - Ring
 - Mesh
- Point-to-multipoint
 - Bus
 - Ring
 - Star

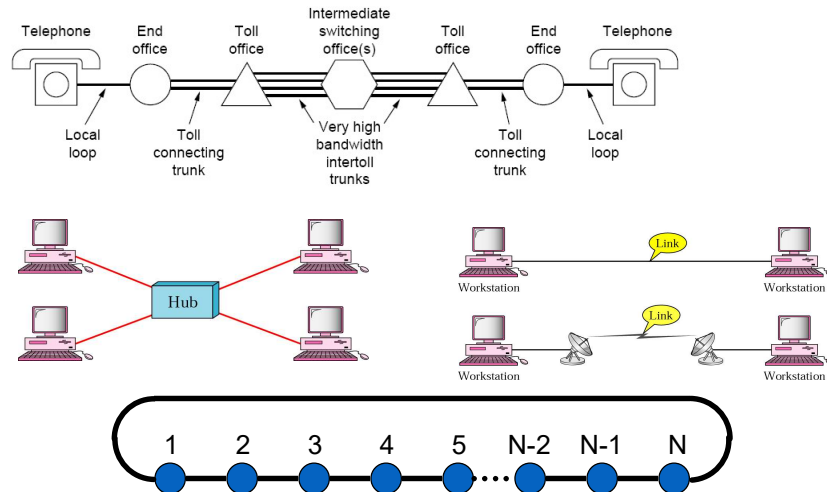


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Point -to-Point



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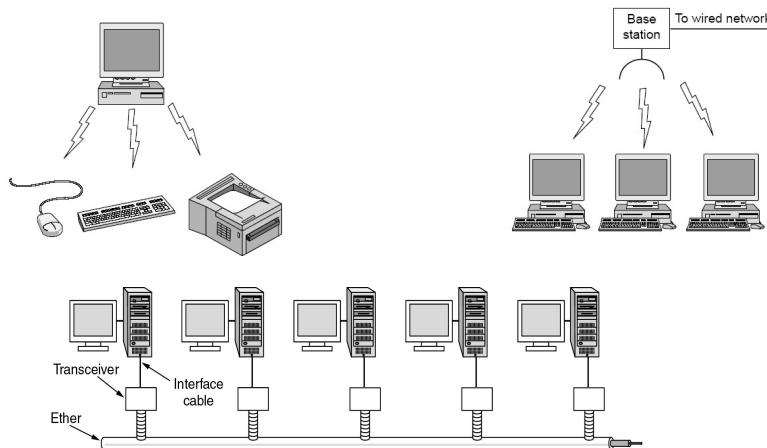
Point-to-point

In point-to-point topology, a transmission line connects two devices.

- Link between two devices:
 - 1 line (half duplex) or
 - 2 lines (duplex)
- *In case of half duplex transmission, there may be collision if two devices on the same link send data at the same time*

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Point-to-multipoint



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Point-to-multipoint

- Common character of point-to-multipoint topo is to use a unique medium to connects multiple nodes.
- Data is broadcasting over the medium
- Collision when two nodes transmit signal in the same time
- Need a control mechanism to allow a single node to transmit → *multiple access method* → see in Datalink layer.

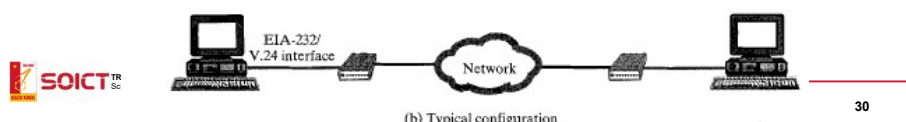
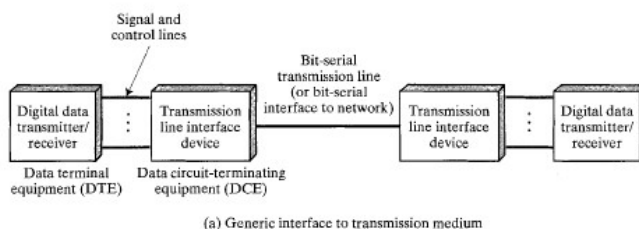


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Medium interface

- Data terminal equipment (DTE)
 - Have data to transmit but has no feature for transmission
 - Need an additional device for accessing the media
- Data circuit terminating equipment (DCE)
 - transmit bits on the media
 - transmit data and control information to another DCE
- Need a clear interface standard between DTE - DCE



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Medium interface

The medium interface is defined according to different aspects:

- Mechanism
 - Define the form of the interface, number of pins for assuring the interfaces match together
- Electrics
 - Define the level of voltage to be used
 - Define the length of pulse (frequency)
 - Define encoding method
- Functionalities
 - Functionality of each pins
 - There are 4 groups of pins: data, control, synchronization, ground
- Procedure
 - Lists of events to perform for transmitting data

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Example: EIA-232-E/RS-232

- Define for serial communication
- Mechanism: ISO 2110
- Electrics: V. 28
- Functionality: V. 24
- Procedure: V. 24



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Example: V.24 /EIA-232-E

- Mechanic:
 - 25 or 15 pins
 - Transmission distance 15m
- Electrics
 - Digital data
 - 1=-3v, 0=+3v (NRZ-L)
 - Data rate 20kbps
 - Transmission distance < 15m

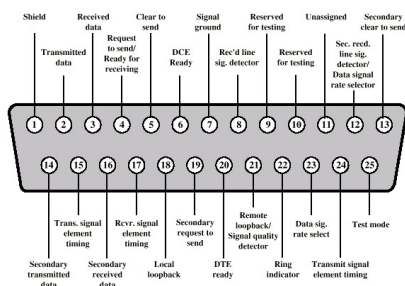


Figure 6.5 Pin Assignments for V.24/EIA-232 (DTE Connector Face)



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Encoding Technics: Data to Signals representations

Introduction
 Digital data - digital signal
 Digital data - analogical signal
 Analogical data - digital signal
 Analogical data - analogical signal
 Reading Chapter 5, Data and Computer Communications, William Stallings

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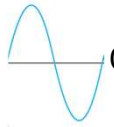
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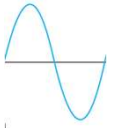
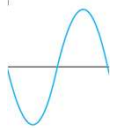
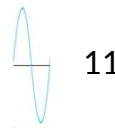
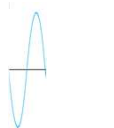
Basic ideas

- Representing data by symbols
- A symbol is an elementary signal:

● Set of mapping is the Encoding scheme

Ex: 1: , 0:  or 1:  0: 

● Encoding could be performed by bit or by a group of bits e.g., 2 or 4 or 8 bits.

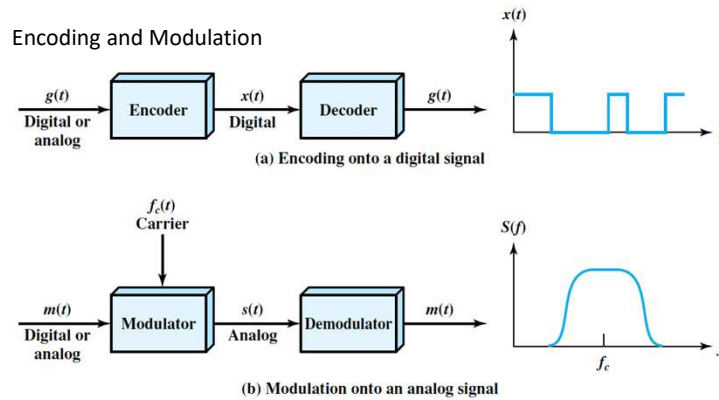
Ex: 00:  01:  10:  11: 

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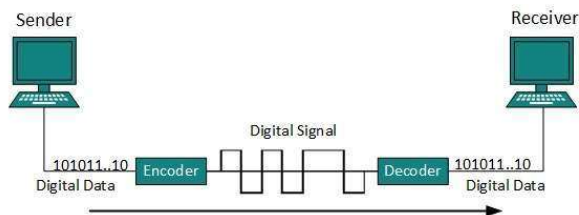
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Basic ideas



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Line coding: represent digital data by digital signals

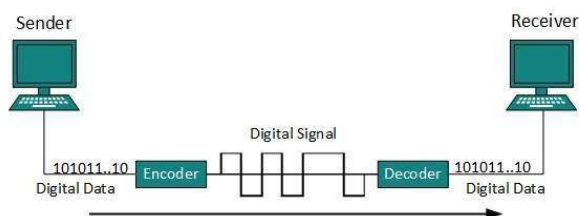


- Use different digital signals at different voltage levels to represent digital data (bits 0 and 1)
 - "A digital signal is one in which the signal intensity maintains a constant level for some period of time and then abruptly changes to another constant level"
- (Data and Computer Communications, 8th Edition, William Stallings)
- An element signal / A symbol/ A pulse of signal
 - Case of digital signal: the shortest signal maintaining a constant level
- Senders encodes a flow of bits → flow of symbols
- Receiver: picks signals, detects symbols, decodes to bits

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Line coding: represent digital data by digital signals

- Receiver should know when a symbol starts or ends: clock synchronization between sender and receiver
- There are many ways to represent 0 and 1 → different encoding methods



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Some line codes

- NRZ
 - NRZ-L, NRZI
- Bipolar
 - Bipolar alternate mark inversion
 - Pseudoternary
- Biphase encoding
 - Manchester
 - Differential Manchester



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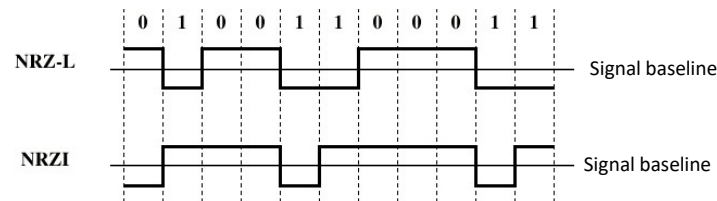
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NRZ-L Non Return to Zero Level

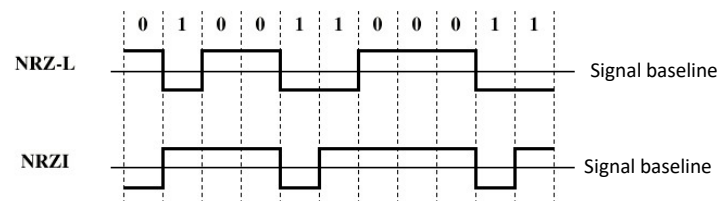
- During bit time, signal does not go back to 0 level
- Signal level is not changed during bit time.
- NRZ-L Non return to zero level
 - Bit 1 signal is in low/high level
 - Bit 0 signal is in high/low level
- 1 bit → 1 symbol



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NRZ-I Non return to zero invert

- Bit 0: signal level is not changed at the beginning of bit time
- Bit 1: signal level is changed at the beginning of bit time
- A differential encoding method :
 - 0 and 1 are represented by the signal level change, not by the level itself.
 - Reliable/ simple.



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Line code consideration

- Three aspects should be considered in any encoding method:
 - **Data speed:** number of bits could be sent per second given symbol rate
 - **Clock recovery on receiver side:** If the clock recovery is correctly at receiver side, the received signal can be incorrectly separated into symbols → cause error in the received data.
 - **DC-component:** Directed Current voltage component.
 - DC-component makes receptor mistakenly detect level of signal (by wrongly detect signal baseline)
 - Encoding should avoid DC-component by maintaining signal mean amplitude around 0.



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NRZ

- NRZ advantages
 - Simple, utilise the maximum capacity of the line
- NRZ weaknesses
 - NRZ does not contain factors supporting clock synchorization
 - Example: when sending a flow of 20 bits 1s, the signal maintains one level and can be detected wrongly at receivers as 19 bits 1s.
 - Contain DC-component when sending a chain of 1s.
- Application
 - Encoding data on magnetic storage
 - Not popular in data transmission



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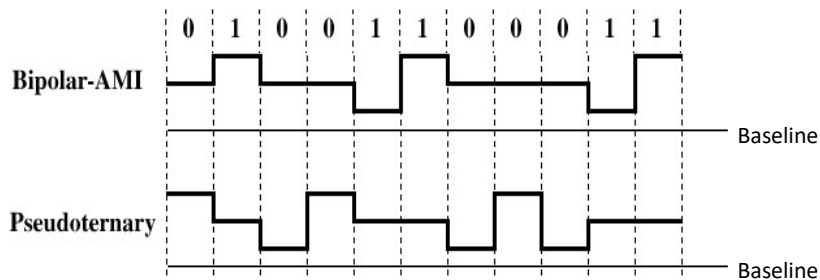
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Bipolar AMI

- Use more than 2 signal levels for 1 bit
- Bipolar alternate mark inversion
 - 0 : No signal
 - 1: Presence of signal. Two consequent 1s have two different signal levels
- pseudoternary
 - 1 : No signal
 - 0 : Presence of signal. Two consequent 0 have two different signal levels



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Bipolar AMI

- No DC component
 - Mean of signal level is always 0
- Good synchronization when sending many bit 1s
- Lost of synchronization when sending many bit 0s
- 3 possible signal levels for 1 bit:
 - Not optimal in transmission line usage.
 - Receiver needs to distinguish 3 levels of signal



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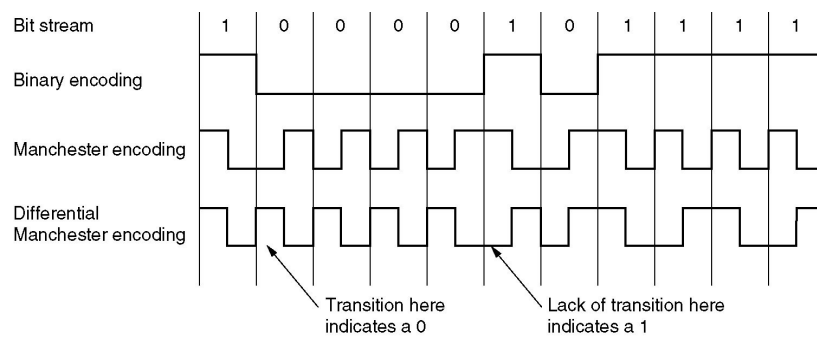
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Biphase: Manchester

- Manchester: Always change signal level in the middle of bit time
 - Bit 1: Signal change from low level to high level.
 - Bit 0: Signal change from high level to low level
- Level change provides a synchronisation mechanism.
- Differential Manchester:
 - 0: signal level change at the beginning of bit
 - 1: **no** signal level change at the beginning of bit
 - Always change signal level in the middle of bit time for synchronization purpose

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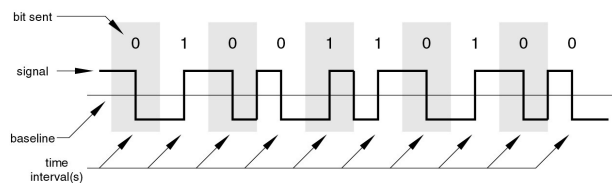
Manchester encoding



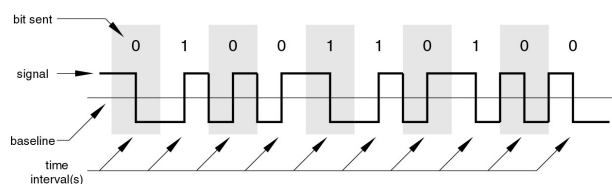
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Manchester encoding

Manchester Encoding



Differential Manchester Encoding



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Units in transmitting digital data in digital transmission

Term	Units	Definition
Data unit	bit	A single bit, Value 0 or 1
Data rate	bit/s	Rate transmitting bit
Signal element /Symbol	Pulse voltage of constant amplitude	Part of a signal that occupies the shortest interval of a signal code
Symbol rate/pulse rate	Number of symbol/s (baud)	Number of symbols generated in a unit of time



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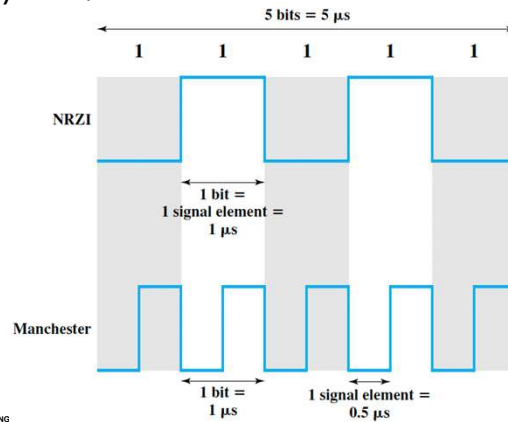
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Symbol rate

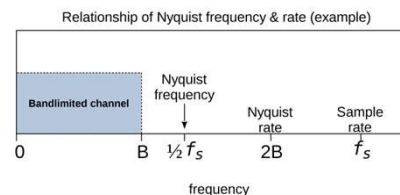
- Number of symbol changes, waveform changes, or signaling events across the transmission medium per unit of time
- Unit: Baud/s = symbol/s



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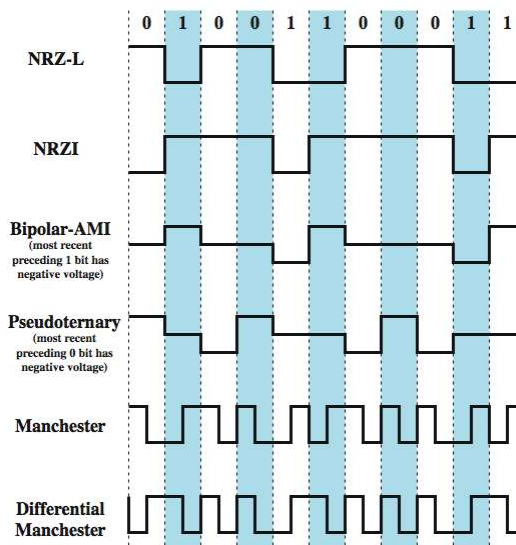
Nyquist rate

- Nyquist rate: 2 times bandwidth of a transmission line
 - Bandwidth: B (Hz)
 - Nyquist rate = 2B (Hz)
- The maximum pulse rate for a baseband channel is the Nyquist rate, and is double the bandwidth
 - Bandwidth B (Hz) → max pulse rate is 2B pulse/s
- Data rate (bps) = pulse rate * number of bits represented by a symbol/pulse.
 - Data rate using NRZ-L?
 - Data rate using Manchester encoding?
- Baseband transmission: signal is directly sent using its original frequency
- Passband transmission: signal is shifted to frequency around carrier frequency.



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Line codes



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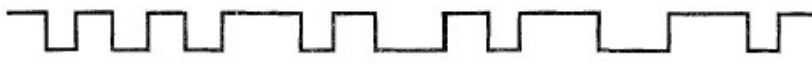
Exercise-01

- Draw the signal generated when we encode following data using different encodings:
 - 11000000 11001101
 - NRZ-L
 - Bipolar AMI
 - Manchester

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Exercise-02

- We know that the following signal is generated from some data using standard Manchester encoding, assume that you act as receiver, please
 - Identify the starting and ending points of signal corresponding to bits
 - Identify the original data sent



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Modulation of digital data to analog signal

- Example: data transmission over telephone line
 - Telephone line accept signal with frequencies between 300Hz - 3400Hz
 - Digital data from the source must be modulated to analog signal to transmit over telephone line.
- 3 modulation techniques:
 - Amplitude-Shift Key
 - Frequency-Shift Key
 - Phase-Shift Key



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Amplitude-Shift Key (ASK)

- Varying the amplitude of the carrier wave to represent digital data

$$s(t) = \begin{cases} A \cos(2\pi ft) & \text{for } 1 \\ 0 & \text{for } 0 \end{cases}$$

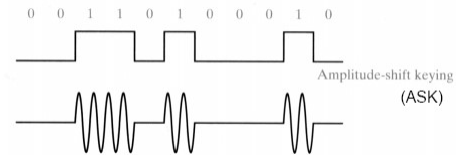
- Carrier original wave function:

$$A \cos(2\pi ft)$$

- 0 and 1 are represented by signals of two different amplitudes. Ex:

- Amplitude A for 1
- Amplitude 0 for 0

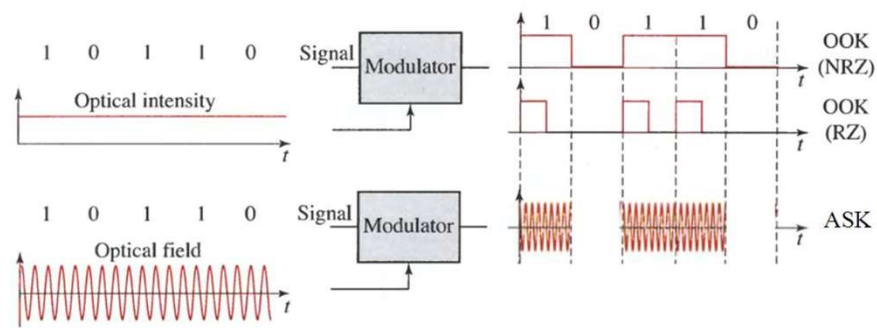
- Usually used in optical fiber



On-Off Keying (OOK)

- Used in optical fiber
- A kind of ASK.
 - 1: having light during the bit time (turn on the light source).
 - 0: no light during the bit time (turn off light source).
- OOK may be realized in two forms:
 - NRZ: light is on during all bit time of bit 1.
 - RZ (return-to-zero): light is on only in a part of 1.

On-Off Keying (OOK)



On off key observed from optical intensity (upper figure)
And optical field (lower figure)

Frequency-shift key (FSK)

- Varying the frequency of the carrier wave to represent digital data
- Two symbols to represent bit 0 and 1 are two carrier signals of different frequencies.
- Lower error rate
- Used for transmitting data over telephone line (low frequency) or wireless network (high frequency)

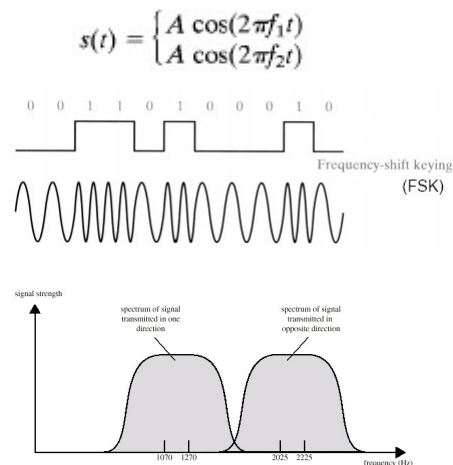
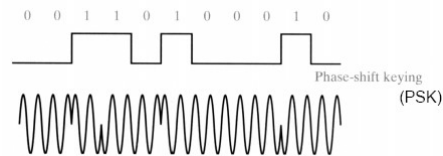


Figure 5.8 Full-Duplex FSK Transmission on a Voice-Grade Line

Phase-shift key (PSK)

- Carrier signal at different phases to represent bit 0 and 1
- A symbol is a signal with one phase

$$s(t) = \begin{cases} A \cos(2\pi f_c t + \pi) & \text{binary 1} \\ A \cos(2\pi f_c t) & \text{binary 0} \end{cases}$$



- By using more than 2 phases, more than two symbols can be generated.
 - A symbol may be used to represent more than one bits.
- May be combined with other modulation methods.

$$s(t) = \begin{cases} A \cos(2\pi f_c t + 45^\circ) & 11 \\ A \cos(2\pi f_c t + 135^\circ) & 10 \\ A \cos(2\pi f_c t + 225^\circ) & 00 \\ A \cos(2\pi f_c t + 315^\circ) & 01 \end{cases}$$



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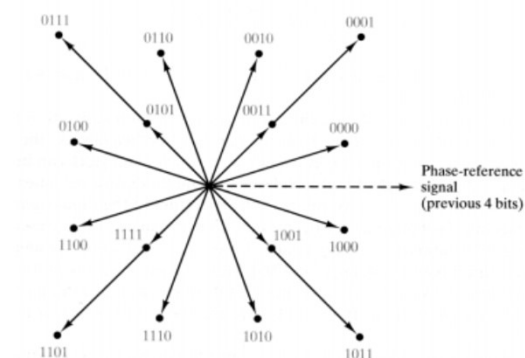
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PSK is used in combination with ASK

Constellation diagram

– 9,600 bps modem (2,400 baud x 4)



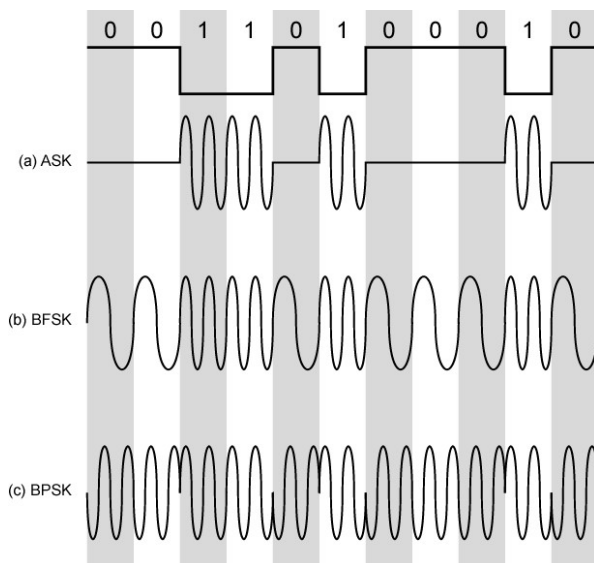
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Digital to analog modulation recapitulation



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Analog signal to digital data

- Used to digitalize real life analog signal, for example voice signal.
- Converting the analog signal to digital data then
 - Encode digital data to digital signal for transmission.
 - Using line codes: NRZ-L, Manchester etc...
- Two methods for converting analog signal to digital data
 - Pulse Code Modulation
 - Delta Modulation



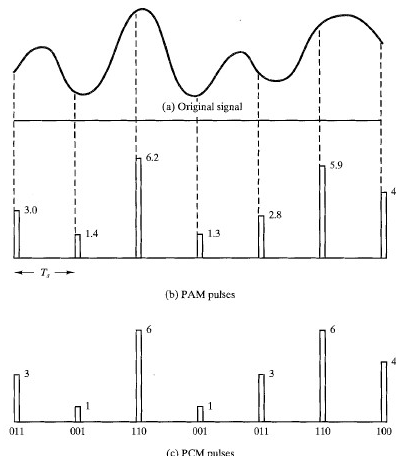
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Pulse Code Modulation (PCM)

- Pulse Code Modulation
- Sample signal according to Nyquist-Shannon sampling theorem
 - If sample a signal with frequency $\geq 2 \times \text{max frequency}$, the original signal can be recovered fidely from the samples.
 - Ex: human voice has greatest frequency of 4300Hz, it should be sampled with at least 8600 sample/s.
- Sample are then quantized
 - Approximate the amplitude of the sample by some digital values
 - 4 bit quantization $\rightarrow$ 16 different possible sample amplitudes.
 - More bits, more precise.



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Delta Modulation

- Weakness of PCM: generate a lot of data $\rightarrow$
- Delta modulation: Use a single bit for each sample
- Idea: Instead sending the real value of the sample $x(t)$, send the variation in sample value in comparison with the last sample: approximative value $\hat{x}(t)$.
 - If $x(t) > \hat{x}(t)$: send bit 1
 - If $x(t) < \hat{x}(t)$: send bit 0



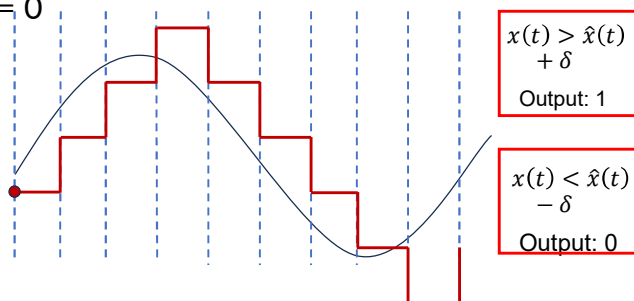
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Delta Modulation

- Nếu $x(t) > \hat{x}(t) \rightarrow \hat{x}(t) := \hat{x}(t) + \delta$
 - Output = 1
- Nếu $x(t) < \hat{x}(t) \rightarrow \hat{x}(t) := \hat{x}(t) - \delta$
 - Output = 0



Transmitting signal: 1 1 1 0 0 0 0 1 1



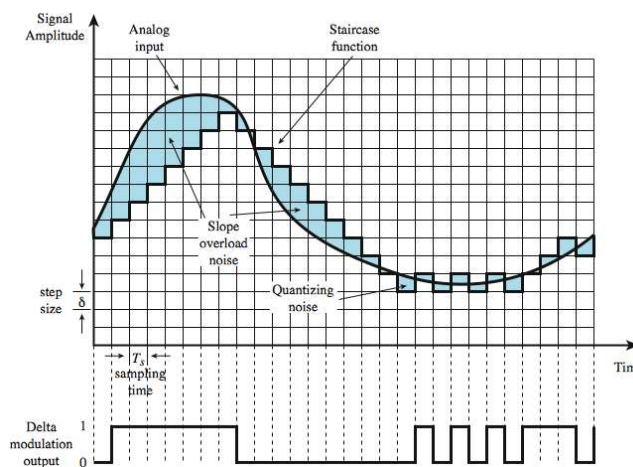
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Delta Modulation

- Parameters of the modulation
 - Step delta
 - Sampling rate
- Errors:
 - When the signal varies slowly: quantizing noise
 - When the signal varies quickly: overloaded noise



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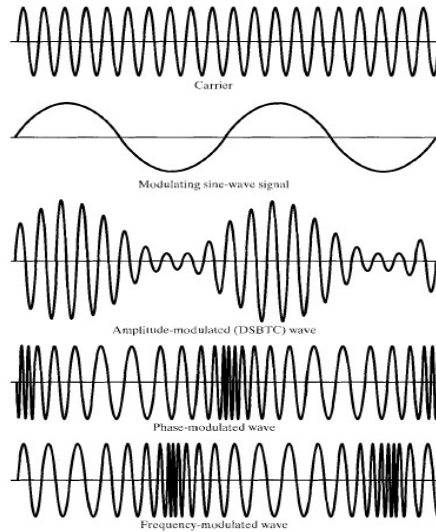
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Analog data – analog signal

- Send analog data over a carrier using analog signal
- Integrate signal $m(t)$ to carrier wave with F_c frequency to a signal with frequency around F_c
- Technic is used to shift signal to a frequency appropriate to the frequency of the transmission channel
- 3 methods
 - Amplitude modulation
 - Frequency modulation
 - Phase modulation



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